Bioinformatics algorithms

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Sequence alignment – exercises

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1. **Global sequence alignment 1**

   Compute the global sequence alignment of the following two strings "DESPERATE" and "ASPARAGUS", using these score parameters:

   - Gap penalty: -2
   - Mismatch penalty: -1
   - Match score: 1

   Answer the following questions:

   (a) What is the final alignment score?

   (b) Are there multiple equivalent solutions?

2. **Longest common subsequence (LCS)**

   Determine the length of the longest common subsequence of the two strings from the first exercise. You can modify the previous exercise in a very simple way to find the answer ...

3. **Global sequence alignment 2**

   Compute the global sequence alignment of the following two DNA sequences "AGCTTAGTTA" and "CGTTAGTAG", using these score parameters:

   - Gap penalty: -2
   - Mismatch penalty: -1
   - Match score: 1

   Answer the following questions:

   (a) What is the final alignment score?

   (b) Are there multiple equivalent solutions?

   (c) Verify that the final score you calculated is correct by inferring it also directly from the alignment which you obtained (helps to cross-check and identify potential errors in the procedure)!

   (d) Does it make any difference if you use a gap penalty of -1 instead (same as mismatch penalty)? Compute the alignment again and compare the results. Can you explain the differences you observe (if any)?

4. ***Modified* global sequence alignment**

   How could you modify the global sequence alignment, such that gaps (insertions and deletions) at the beginning and the end of the alignment have no penalty (gap penalty of 0 only at the beginning and end of the alignment)?

   Apply your proposed approach to the following problem:

   Find the best global alignment for the two sequences "TCGTAC" and "GCTCG" such that,

   - match score = 1
   - mismatch penalty = -1
   - gap penalty = -2 (within the alignment)
   - gap penalty = 0 (at the beginning and end of the alignment)

   No need to write any formula, just apply your idea to the problem while you fill the scoring matrix!